

📍 Cotonou, Bénin ✉ perrierosas@gmail.com 🔗 rosasbehoundja.github.io in [rosasbehoundja](#)
🌐 [rosasbehoundja](#)

Artificial Intelligence student passionate about using combinatorial optimization and machine learning to build practical tools that solve real-world challenges across Africa.

Education

IFRI, University of Abomey-Calavi, Bénin Nov 2023 - Present
Bachelor of Science in Artificial Intelligence

Experience

Applied Machine Learning Mentor Bénin
Sèmè City July 2026

- Mentored the top 20 participants of the National Olympiad of Artificial Intelligence (NOAI) 2026, providing guidance in Machine Learning and supporting them throughout their technical challenges.

Research Intern Bénin
Computer Science and Applications Research Laboratory of IFRI (LRSIA) March 2026 – June 2026

- Research on sequence variables, a constraint programming computational domain for routing and sequencing problems; contributions to VRP projects at the lab.

AI Developer Bénin
IFRI & Tekbot Robotics June 2025 – Jan 2026

- Developed and deployed an AI vision model and robotic controls for an automated waste-sorting system, allowing a robot arm to instantly recognize and sort trash in real time to improve urban recycling.

Machine Learning Engineer Intern South Africa
eTihuku June 2025 – Aug 2025

- Worked on SentiMaster an AI platform that collects customer feedback from social media and phone calls, automatically translates and analyzes the hidden emotions and topics, and helps companies predict customer churn.

Selected Projects

- **MPVRP-CC** 🔗: Proposed a variant of the multi-product vehicle routing problem with production changeover costs, modeling distribution with maintenance costs.
- **MDVRP** 🔗: Designed and implemented optimization algorithms (ALNS, PSO, Simulated Annealing, Linear Programming) for multi-depot vehicle routing (MDVRP) under capacity and distance constraints.
- **Opti'plan**: Co-built an automated scheduling model that reduced the university's thesis defense planning time from 3 days to under 5 minutes using Constraint programming.
- **tiny-language-model** 🔗: Built a custom AI language model from scratch, trained to understand and generate French text.
- **ifri-mini-ml-lib** 🔗: Co-built an open-source Python library that reimplements classic AI algorithms from scratch for educational purposes; published on PyPI and used by IFRI students.

Awards & Engagements

1st Prize – IndabaX Benin Republic Hackathon 2024 [Certificate](#) 🔗

2nd Place – Tekbot Robotics Challenge 2025 [Link](#) 🔗

THE LAB & FRIARE Africa: Co-lead an AI awareness community in Bénin; participate in awareness initiatives on AI ethics, transparency, inclusion.

Skills

Programming: Python, Java, SQL, R

AI: Machine learning, Constraints programming, NLP, Metaheuristics

Infra & Tools: PyTorch, Scikit-learn, Gurobi, Azure, ROS2, FastAPI, PyPI/CI-CD

Soft skills: Teamwork, leadership, communication, problem solving, critical thinking.

Languages: French (native) · English (professional)